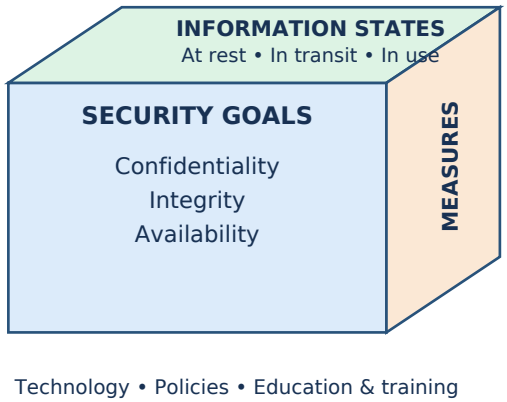
Security Principles of Cybersecurity – McCumber Cube

1. Introduction

The McCumber Cube is a three-dimensional model used in cybersecurity to explain how to protect information. It was developed by John McCumber. It combines three important dimensions to ensure information security.

2. Diagram of McCumber Cube



The diagram shows the three dimensions that work together to protect information.

3. Three dimensions of the McCumber Cube

Dimension	Elements	Explanation
1. Security goals (CIA Triad)	Confidentiality Integrity Availability	Confidentiality: only authorized people can access information. Integrity: information remains accurate and is not changed without authorization. Availability: information and systems are accessible when needed.
2. Information states	Storage (at rest) Transmission (in transit) Processing (in use)	Storage: data saved on a computer or hard disk. Transmission: data transferred through a network or the internet. Processing: data being used by a computer or application.
3. Security measures	Technology Policies and procedures Education and training	Technology: firewalls, antivirus software and encryption. Policies: security rules, access controls and backup policies. Education: teaching users to recognize threats and follow safe practices.

4. Importance of the McCumber Cube

• Helps identify security risks.

• Protects information from unauthorized access and modification.

• Ensures information availability.

• Helps organizations develop complete cybersecurity strategies.

Conclusion: The McCumber Cube provides a systematic approach to cybersecurity by combining security goals, information states and security measures to protect information effectively.